K.R. MANGALAM UNIVERSITY

Database Management Assignment File (ENCS205)

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

Divesh (2401010077)

B. Tech CSE (Core) – 4th Semester

Section: A

Submitted To:

MS. Mansi Kajal

School of Engineering & Technology
K. R. MANGALAM UNIVERSITY
Sohna, Haryana 122103, India

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LAB SHEET: 2

Objective:

To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

```
CREATE DATABASE IF NOT EXISTS DBMS_LAB;
USE DBMS_LAB;
CREATE TABLE CLASS (
   id INT PRIMARY KEY,
   name VARCHAR(50)
);

INSERT INTO CLASS VALUES (1,'Bravo');
INSERT INTO CLASS VALUES (2,'Alex');
INSERT INTO CLASS VALUES (4,'Cheng');
```

2. How do you create a table named EMPLOYEES with the specified attributes?

```
CREATE TABLE EMPLOYEES (
   Employee_Id INT PRIMARY KEY,
   First_Name VARCHAR(50),
   Last_Name VARCHAR(50),
   Email VARCHAR(100),
   Phone_Number VARCHAR(20),
   Hire_Date DATE,
   Job_Id VARCHAR(20),
   Salary DECIMAL(10,2),
   Commission_Pct DECIMAL(5,2),
   Manager_Id INT,
   Department_Id INT
);

INSERT INTO EMPLOYEES VALUES
(101,'John','Austin','john@gmail.com','9999999999',CURDATE(),'IT_PROG',5000,NULL,10
0,60),
(102,'Sara','Smith','sara@gmail.com','8888888888',CURDATE(),'HR_REP',4500,NULL,100,7
0),
(103,'David','Lee','david@gmail.com','7777777777',CURDATE(),'IT_PROG',4800,NULL,10
1,80),
(104,'Chris','Martin','chris@gmail.com','6666666666',CURDATE(),'SA_REP',6000,NULL,10
0,60);
```

3. How do you create a table named Client_master with the specified attributes?

```
CREATE TABLE Client_master (
   ClientNO VARCHAR(10) PRIMARY KEY,
   Name VARCHAR(50),
   Address VARCHAR(100),
   City VARCHAR(50),
   State VARCHAR(50),
   Bal_due DECIMAL(10,2)
);
```

4. What SQL statement would you use to rename the table Client_master to Client12?

```
ALTER TABLE Client_master RENAME TO Client12;
```

5. How do you display the bal_due column heading as “BALANCE”?

```
SELECT Bal_due AS "BALANCE"
FROM Client_master;
```

Part A: Transactions and Savepoints (CLASS Table)

```
SET autocommit = 0;
START TRANSACTION;

INSERT INTO CLASS VALUES (5,'Rahul');

UPDATE CLASS SET name='Abhijeet' WHERE id=5;
SAVEPOINT A;

INSERT INTO CLASS VALUES (6,'Chris');
SAVEPOINT B;

INSERT INTO CLASS VALUES (7,'Bravo');
SAVEPOINT C;

SELECT * FROM CLASS;

ROLLBACK TO B;
SELECT * FROM CLASS;

ROLLBACK TO A;
SELECT * FROM CLASS;
```

OUTPUT:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Co

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Co

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Co

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
NULL	NULL

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.

```
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Cor

Employee_Id	First_Name	Last_Name	Salary
101	John	Austin	5000.00
102	Sara	Smith	4500.00
103	David	Lee	4800.00
104	Chris	Martin	6000.00
NULL	NULL	NULL	NULL

2. List out the employees who work under manager 100.

```
SELECT *
FROM EMPLOYEES
WHERE Manager_Id = 100;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	John	Austin	john@gmail.com	9999999999	2026-02-28	IT_PROG	5000.00	NULL	100	60
102	Sara	Smith	sara@gmail.com	8888888888	2026-02-28	HR_REP	4500.00	NULL	100	70
104	Chris	Martin	chris@gmail.com	6666666666	2026-02-28	SA_REP	6000.00	NULL	100	60
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Find the names of the employees who have a salary greater than or equal to 4800.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_Name	Last_Name
John	Austin
Sara	Smith
David	Lee
Chris	Martin

4. List out the employees whose last name is ‘AUSTIN’.

```
SELECT *
FROM EMPLOYEES
WHERE Last_Name = 'Austin';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	John	Austin	john@gmail.com	9999999999	2026-02-28	IT_PROG	5000.00	NULL	100	60
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. Find the names of the employees who work in departments 60, 70, and 80.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60,70,80);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_Name	Last_Name
John	Austin
David	Lee
Chris	Martin

6. Display the unique Manager_Id.

```
SELECT DISTINCT Manager_Id
FROM EMPLOYEES;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Manager_Id
100
101

Part C: Updating Records (Client_master Table)

1. Insert five records into the Client_master table.

```
INSERT INTO Client_master VALUES
('C101','Rohan','MG Road','Delhi','Delhi',4500),
('C102','Aman','Park Street','Kolkata','WB',6000),
('C123','Priya','Sector 10','Gurgaon','Haryana',4800),
('C104','Neha','Ring Road','Mumbai','Maharashtra',7000),
('C105','Vikas','Main Market','Jaipur','Rajasthan',3000);

COMMIT;
```

2. Find the names of clients whose bal_due > 5000.

```
SELECT Name
FROM Client_master
WHERE Bal_due > 5000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name
Aman
Neha

3. Change the bal_due of ClientNO "C123" to Rs. 5100.

```
UPDATE Client_master
SET Bal_due = 5100
WHERE ClientNO = 'C123';

COMMIT;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

BALANCE
4500.00
6000.00
7000.00
3000.00
5100.00